



IMPLEMENTATION OF THE UNESCO 2021 RECOMMENDATION ON OPEN SCIENCE IN MALAYSIA

December 2024

Country	Malaysia
Organisation	
Name	Academy of Sciences Malaysia
Website	https://www.akademisains.gov.my/
Email address	
Phone number	
Please describe the role/mandate of your organization:	Thought leader in science, technology, innovation and economy (STIE) for Malaysia. Supporting the government through evidence-based policymaking.
Officially designated contact	
Full name	Professor Dr Shaliza Ibrahim FASc
Position	Fellow of the Academy of Sciences Malaysia, Co-Chair of Malaysia Open Science Alliance
Email address	

Brief description of the consultation process established for the preparation of the report

The survey was disseminated to ministries, higher learning institutes, research agencies and open science enthusiasts to collect information on open science activities that have been conducted in Malaysia. The survey was conducted from 7th October to 9th November 2024. The finding of the report is endorsed by the Malaysia Open Science Alliance, the Council members of the Academy of Sciences Malaysia and the International Division, Ministry of Science, Technology and Innovation.

Other organization(s) or entity(ies) including non-governmental, consulted for completing this survey:

No.	Name of the organisation	Website	E-mail address	Sector
1.	Universiti Malaysia Terengganu	https://www.umat.edu.my/		Public
2.	Universiti Putra Malaysia	www.upm.edu.my		Public
3.	Sultanah Bahiyah Library, Universiti Utara Malaysia	https://library.uum.edu.my/		Public
4.	Institut Kefahaman Islam Malaysia (Institute of Islamic Understanding Malaysia)	https://www.ikim.gov.my/		Public
5.	Universiti Malaya	https://www.um.edu.my/		Public
6.	Department of Chemistry Malaysia	www.kimia.gov.my		Public
7.	Forest Research Institute Malaysia	https://www.frim.gov.my/		Public
8.	Department of Wildlife and National Park	https://www.wildlife.gov.my		Public
9.	Tuanku Bainun Library, Universiti Pendidikan Sultan Idris	https://pustaka.upsi.edu.my		Public
10.	Laman Hikmah Library, Universiti Teknikal Malaysia Melaka (UTeM)	https://library.utm.edu.my		Public
11.	Institute for Medical Research			Public
12.	Asia Pacific University of Technology & Innovation	https://www.apu.edu.my/		Public
13.	Ministry of Housing and Local Government	https://www.kpkt.gov.my/		Public
14.	Jabatan Perkhidmatan Awam (Public Service Department)	www.jpa.gov.my		Public

15.	Ministry of Works Malaysia	www.kkr.gov.my		Public
16.	Ministry Of Women, Family and Community Development	www.kpwkm.gov.my		Public
17.	Infrastructure University Kuala Lumpur	www.iukl.edu.my		Public
18.	Malaysian Pepper Board	www.mpb.gov.my		Public
19.	Ministry of Human Resources	https://www.mohr.gov.my/		Public
20.	Malaysian Technology Development Corporation Sdn Bhd	http://mtdc.com.my		Public
21.	Royal College of Surgeons in Ireland and University College Dublin (RCSI & UCD) Malaysia Campus	https://www.rumc.edu.my/		Public
22.	Ministry Energy Transition and Water	www.petra.gov.my		Public
23.	Jabatan Perhilitan Negeri Perak	www.wildlife.gov.my		Public
24.	Universiti Malaya-Wales	https://www.umwales.edu.my/		Public
25.	Malaysian Palm Oil Board	www.mpob.gov.my		Public
26.	Institute For Youth Research Malaysia (IYRES)	https://www.iyres.gov.my/		Public
27.	Malaysian Cocoa Board	http:///koko.gov.my		Public
28.	Ministry of Plantation and Commodities	www.kpk.gov.my		Public
29.	National Institutes of Health Malaysia	www.nih.gov.my		Public
30.	Universiti Malaysia Sarawak (UNIMAS)	www.unimas.my		Public
31.	Universiti Tunku Abdul Rahman	www.utar.edu.my		Public

32.	Universiti Kebangsaan Malaysia	https://www.ukm.my/portalukm		Public
33.	Malaysia Board of Technologists	https://www.mbot.org.my/		Public
34.	Spectrum International University College	www.siuc.edu.my		Public
35.	Universiti Pendidikan Sultan Idris	www.upsi.edu.my		Public
36.	Institute for Clinical Research	icr.nih.gov.my		Public
37.	Ministry of Agriculture and Food Security	kpk.m.gov.my		Public
38.	Institute for Health Behavioural Research (IHBR)	https://iptk.moh.gov.my/		Public
39.	Institute for Health Management, National Institutes of Health	ihm.moh.gov.my		Public

SECTION 1

PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1. Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

In Malaysia, the Recommendation is promoted through various ministries, agencies and higher learning institutes. The Ministry of Science, Technology and Innovation in collaboration with the Academy of Sciences Malaysia launched the Malaysia Open Science Platform in 2023. This platform aims to collect and consolidate Malaysian raw research data in one platform to enhance transparency and accessibility to research data. Other active advocator of open science and the Recommendation are, but not limited to:

Ministries: Ministry of Higher Education, Ministry of Housing and Local Government, Ministry of Communication, Ministry of Digital, Ministry of Natural Resources and Environmental Sustainability, Ministry of Agriculture and Food Security, Ministry of Health, Ministry of Education

Agencies: National Institutes of Health, Federal Land Development Authority, National Digital Department, Forest Research Institute Malaysia, MIMOS Sdn Bhd, Department of Agriculture

Institutes of Higher Learning:

Public - Universiti Malaya, Universiti Sains Malaysia, Universiti Teknologi Malaysia, Universiti Kebangsaan Malaysia, Universiti Putra Malaysia, Universiti Malaysia Sabah, Universiti Malaysia Sarawak, Universiti Tun Hussein Onn Malaysia, Universiti Malaysia Terengganu

Private - Sunway University, UCSI University, International Medical University, Universiti Tunku Abdul Rahman

International network: International Science, Technology and Innovation Centre for South-South Cooperation under the auspices of UNESCO (ISTIC) and the Science Technology Engineering Innovation Policy Asia and the Pacific Network (STEPAN)

The dissemination of information should however be expanded to more ministries, higher learning institutes, research institutes and the public, especially from the social science and humanities field.

2. Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

Yes. The Recommendation is available in English, but related

information is also disseminated in Bahasa Malaysia.

3. Have there been awareness raising activities on open science, including all its key elements, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities/entities?

1. Symposium on Open Science
2. Open Science Talks and Webinars
3. Malaysia Open Science Platform initiative (<https://mosp.gov.my/>)
4. Data Stewardship on Open Science training
5. Podcasts (<https://www.akademisains.gov.my/asm-podcast/>)
 - a. Evolution of Librarian Role in Data-Driven Society
 - b. How does the field of medicine support research data sharing
6. Launching of data repositories
7. Awareness in natural resources and protected area (E.g. Save the Freshwater Fish, Save the Malayan Tiger)
8. Enhancing and Strengthening Regional Open Science Implementation Mechanism Workshop (<https://tinyurl.com/yc2h5x45>)
9. Roadshows and exhibitions
10. Community Engagement in the National Park
11. Collaboration with local universities
12. Open science activities by universities
 - a. <https://cdo.unimas.my/open-asia/activities/open-science>
 - b. <https://openscience.ukm.my/home>
 - c. <https://psnz.umt.edu.my/news-and-events/>
 - d. <https://lib.usm.my/index.php/academicians-researchers/open-science>

4. Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science, in publicly funded research?

National level

1. National Policy on Science, Technology and Innovation 2021-2030 specified on encouraging open data sharing to drive national innovation. ([Dasar Sains, Teknologi dan Inovasi Negara \(DSTIN\) - Portal Rasmi Kementerian Sains, Teknologi Dan Inovasi](#))
2. Formation of the Malaysia Open Science Alliance, under the Academy of Sciences Malaysia involving ministries, agencies, institutes of higher learning and research institutes The National Guidelines on Open Science for Public Funded Research was developed in 2022 by the Academy of Sciences Malaysia. These guidelines have been accepted as a subset under the National Data Sharing Policy in 2023. (<https://mosp.gov.my/announcement/document/18/guidelines-of-open-science-for-public-funded-research>)
3. The Malaysia Open Science Platform (MOSP) is launched to share research

- data especially from the public funded research. The platform is connected to institutional repositories and allows access of research data to the public.
4. Providing various awareness activities.
 5. The National Digital Library is a national knowledge infrastructure by the Ministry of Higher Education.
 6. Public funding bodies such as the Ministry of Higher Education (MOHE) have included open science policies in their funding requirements.

Institutional Level

1. Development of institutional data repositories.
2. Implementation of institutional Research Data Management Policy and/or Data Management Plan. For example:
 - a. Universiti Malaya - <https://openscience.um.edu.my/um-research-policy>
 - b. Universiti Malaysia Sarawak - <https://cdo.unimas.my/unimas-dataverse/policies-and-guidelines>
3. The formation of open science committees in the institution to drive open science.
4. Including article processing charges for open access journals in research grants.
5. Promote multi-stakeholder collaboration and contribution for dedicated open science funding.
6. To establish a sustainable model for an identifiable, discoverable, accessible, and shared open science technology and infrastructure.
7. To enable a participatory/co-creation approach (which includes partnership design, implementation, monitoring, evaluation and reporting) for transparency and integrity among the participating members.
8. To develop a sustainable capacity building for open science through learning and training modules and a co-designed approach for open science actors. Currently, capacity building on open science is being organized in institutes focusing on data management, open-access publishing, and the use of open research infrastructures to promote a culture of openness.
9. Incorporate diversity and inclusiveness (the core values of open science) of knowledge and knowledge systems in every aspect of open science implementation.
10. Adoption of policies to make research outputs from publicly funded projects openly accessible.
11. Actively involved in collaborative projects with other local and international institutions, promoting open science through shared research infrastructures. This includes participation in regional open data platforms, encouraging researchers to share data for the collective benefit of scientific progress.
12. Promoting the use of open repositories and metadata standards to ensure that research data can be effectively shared and reused by the global research community.

5. Have your country/organization conducted or planned any open science

activities?

Yes. Mainly in open science in scientific publication, open research data and open educational resources. In addition to that, there are also planning of activities relating to open dialogue with local communities, citizen and participatory science, physical open infrastructure and open dialogue with indigenous people. Generally, the activities are categorised as below:

1. Development of research data management policy in higher learning institutes
2. Training on data stewardship for open science
3. Engagements with researchers for open research and data repositories for sharing research data
4. Journal of wildlife and parks
5. Malaysia Biodiversity Information System – platform to share specimens of biodiversity in Malaysia
6. Dialogues about conservation of wildlife habitat

Activities by institutes

1. Universiti Sains Malaysia: <https://lib.usm.my/index.php/academicians-researchers/rdm-services>
2. Universiti Malaysia Terengganu: <https://psnz.umt.edu.my/news-and-events/>
3. Universiti Malaya:
 - a. Launch of Universiti Malaya Open Science (UMOS):
<https://openscience.um.edu.my/>
 - b. Scientific Publication:
<https://umlibguides.um.edu.my/c.php?g=395658&p=2688786>
 - c. Open Research Data: <https://researchdata.um.edu.my/>
 - d. Open Educational Resources: <https://ocw.um.edu.my/>
 - e. Citizen and Participatory Science:
<https://sustainability.um.edu.my/news/um-celebrates-citizen-science-month-in-empowering-sdgs-literacy>
4. Universiti Putra Malaysia:
 - a. Scientific Publication:
 - i. 1. Webinar: Empowering Research through Open Science: Why This Matters Now More Than Ever
https://www.facebook.com/photo?fbid=1075155141287075&set=a.702897688512824&locale=ms_MY
 - ii. 2. Webinar: Open Science Trends and How They Influence the Impact of Your Journal
https://www.facebook.com/photo/?fbid=1047234107412512&set=a.702897688512824&locale=ms_MY
 - iii. 3. Webinar: Research Data Management - Dive into the Details
https://www.facebook.com/photo/?fbid=1047270580742198&set=a.702897688512824&locale=ms_MY
 - iv. 4. Webinar: Research Data - "Why It Matters and How to Manage It"

https://www.facebook.com/photo/?fbid=5933855986631696&set=a.131787750171911&locale=ms_MY

- v. 5. Webinar: Uncover Details around Research Data Management
 - b. Open Research Data: <http://putrarepo.upm.edu.my/action.do>
 - c. Open Educational Resources: <http://putraoer.upm.edu.my/>
 - d. Open Source Software and Source Code:
https://lib.upm.edu.my/article/use_of_open_source_software_applications_at_sultan_abdul_samad_library-55307
- 5. Universiti Malaysia Sarawak: <https://cdo.unimas.my/open-asia/activities/open-science>
- 6. Universiti Kebangsaan Malaysia:
 - a. Open Educational Resources: <https://www.ukm.my/i-kom/>
- 7. Universiti Tunku Abdul Rahman:
 - a. Engagement with community:
<https://library.utar.edu.my/Databases2-0.php>
<https://www2.utar.edu.my/dssc/index.jsp?fccatid=478&fcontentid=3123&f2ndcontentid=4932>
 - b. Societal engagement: <https://softskill.utar.edu.my/Perkampungan-Orang-Asli.php>
 - c. Fundraising project:
<https://news.utar.edu.my/news/2017/Sept/13/03/03.html>
<https://utarhospital.org.my/be-the-light-in-someones-dark-world/>
- 8. Forest Research Institutes Malaysia
 - a. Scientific Publication – Journal of Tropical Forest Science (JTFS):
<https://jtfs.frim.gov.my/jtfs>
 - b. Open Research Data – RIMS:
<https://info.frim.gov.my/infocenter/rims2/home.cfm>
 - c. Citizen and Participatory Science: a. List of project i. Citizen Science Programme for Conservation of Wetland in FRIM, Selangor ii. Preliminary Study to Develop a Citizen Science Protocol for Bukit Kiara Park Firefly Community and Habitat Assessment iii. Citizen Science Project (KHIND)
 - d. Open Dialogue with Indigenous Peoples: a. Documenting traditional knowledge of Indigenous Peoples from 5 sub ethnics in Peninsular Malaysia
 - e. Open hardware – RIMS:
<https://info.frim.gov.my/infocenter/rims2/home.cfm>

SECTION 2

DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

- 6. Does your country have a national policy, strategy or plan of action on science, technology and innovation (STI)?**
(Title, Authority in Charge, Entities involved and the process of development, Year of Adoption, Year of Last update, Key Areas, Link)

- a. National Science, Technology and Innovation Policy 2021-2030
Ministry of Science, Technology and Innovation
2021
Key areas of science, technology and innovation for the nation
<https://www.mosti.gov.my/en/dstin/>
- b. National Science, Technology and Innovation Policy 2021-2030 Action Plan
Ministry of Science, Technology and Innovation
2021
Key areas of science, technology and innovation for the nation. Six thrusts:
Responsive STI Governance, Technology Development Through R&D&C&I, Local
Technology Based Industry, Adaptive STI Talent, Enculturation and Application of
STIE.
<https://www.mosti.gov.my/wp-content/uploads/2022/03/Action-Plan-of-National-Science-Technology-and-Innovation-Policy-2021-2030.pdf>

7. In your country, are there specific policy instruments that aim to promote open science?

- a. National Guidelines on Open Science for Public Funded Research
Guidelines on research data management and research data sharing. Promote
openness and accessibility.
Ministry of Science, Technology and Innovation with Academy of Sciences
Malaysia
<https://mosp.gov.my/announcement/document/18/guidelines-of-open-science-for-public-funded-research>
2022
- b. General Circular on Open Research Data
Circular and guidelines on the implementation of government open data to
enhance transparency through data sharing.
National Digital Department (previously led by the Department of Prime Minister)
<https://navy.mil.my/images/widgetkit/Penerbitan/Panduan/dataset/Pekeliling%20Am%20Bil%201%202015%20DTSA.pdf>
2015

8. In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation?

- a. National Science, Technology and Innovation Policy 2021-2030 Action Plan
Ministry of Science, Technology and Innovation
2021
Key areas of science, technology and innovation for the nation. Six thrusts:
Responsive STI Governance, Technology Development Through R&D&C&I, Local

Technology Based Industry, Adaptive STI Talent, Enculturation and Application of STIE.

<https://www.mosti.gov.my/wp-content/uploads/2022/03/Action-Plan-of-National-Science-Technology-and-Innovation-Policy-2021-2030.pdf>

- b. National Data Sharing Policy
Ministry of Digital (formerly Ministry of Communication and Digital)

In addition to the national level implementation plan, open science is also considered at the institution level. This is achieved through research data management and open science policies, open science guidelines and data access policies. The implementation plan focuses on fostering open science practice among researchers, ensuring research is handled systematically, enhancing integrity, accountability, long-term availability and sharing. This also complies with the funding agency requirements.

For example, Universiti Malaysia Sarawak (UNIMAS) Open Science Research Data Management Policy aims to promote good practice in open science research data management in the conduct of research projects at UNIMAS and to clarify the responsibilities of both individual researchers and the institution in relation to good open science research data management and support.

9. In your country, have open science practices been integrated in existing research funding mechanisms?

National Level

Malaysian funding agencies, such as the Ministry of Higher Education and the Malaysian Research Institute, often include requirements for open access publication and data sharing in their grant application.

Institutional Level

Some institutes such as the National Institutes of Health (NIH) Malaysia has already included in their funding requirements that the research datasets produced from research funded by the Ministry of Health is to be deposited into their institutional repository. Some embargo period and exception clauses are provided. Some higher learning institutes such as the Universiti Malaysia Terengganu (UMT) have incorporated data sharing requirements that need to be followed and accepted by the recipients of UMT internal funding (DP-UMT).

For both levels however, there are still gaps in funding implementation following the fear of plagiarism and lack of awareness on open science.

10. Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation?

Malaysia has undertaken several initiatives to foster equitable public-private partnerships (PPPs) in open science, aligning with the 2021 UNESCO Recommendation on Open Science. These efforts aim to enhance collaboration between public research institutions and private entities, promoting transparency, accessibility, and innovation in scientific research.

The Malaysia Open Science Alliance (MOSA) is formed by the Academy of Sciences Malaysia. The Alliance involves the ministry, agencies, research institutions and higher learning institutions (public and private). Malaysia is also in collaboration with international open science organization such as Australian Research Data Commons, Japan Research Center for Open Science and Data Platform, African Open Science Platform and Committee on Data (CODATA) under the International Science Council (ISC).

With the intention of moving towards open innovation, the Malaysia Open Science Platform (MOSP) serves as an avenue for collaboration between the public and private sector. Researchers in universities and research institutes in Malaysia is already implementing the practice of collaborating with the industry to propel innovation and accelerate commercialization. Through the research data sharing platform, the industry is now able to connect with researchers efficiently.

Higher learning institutes are also promoting open access in the research ecosystem to increase transparency and accountability for research. Libraries are providing open resources such as the Malaysian Academic Library Union Catalog (MALCat) and Malaysian Academic Library Institutional Repository (MALRep) to share resources among the academia community.

11. Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy

- a. National Science, Technology and Innovation Policy 2021-2030
Ministry of Science, Technology and Innovation
2021
Key areas of science, technology and innovation for the nation
<https://www.mosti.gov.my/en/dstin/>
- b. National Science, Technology and Innovation Policy 2021-2030 Action Plan
Ministry of Science, Technology and Innovation
2021
Key areas of science, technology and innovation for the nation. Six thrusts: Responsive STI Governance, Technology Development Through R&D&C&I, Local Technology Based Industry, Adaptive STI Talent, Enculturation and Application of STIE.
<https://www.mosti.gov.my/wp-content/uploads/2022/03/Action-Plan-of->

- c. National Guidelines on Open Science for Public Funded Research
Guidelines on research data management and research data sharing.
Promote openness and accessibility.
Ministry of Science, Technology and Innovation with Academy of Sciences Malaysia
<https://mosp.gov.my/announcement/document/18/guidelines-of-open-science-for-public-funded-research>
- d. Research, Development, Innovation, Commercialization and Economy (R,D,I,C,E) Roadmap for Malaysia
2023
This strategic roadmap aims to streamline key strategies and ensure a holistic and vibrant research, development, innovation, commercialisation, and economic ecosystem that emphasises demand-driven innovation for market driven delivery systems.

SECTION 3

INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

12. Are there national research and education networks (NRENs) active in the country?

Yes, Malaysia has an active National Research and Education Network (NREN) known as the Malaysian Research and Education Network (MYREN) [<https://www.mohe.gov.my/en/services/research/myren>]. This network has connected all public universities, private universities, all polytechnics and college communities, a few teachings hospitals, providing high-speed connectivity and digital infrastructure.

It plays a key role in supporting Open Science initiatives by facilitating data sharing, virtual collaboration, and access to global research networks. MYREN connects Malaysia's research community with international NRENs, promoting cross-border collaboration and open access to scientific knowledge. These efforts align with national goals to foster inclusive access to research outputs, enabling seamless collaboration among key stakeholders in the academic and scientific ecosystem.

13. Are there national open science infrastructures in your country?

National Level

1. Malaysian Open Science Platform (MOSP)

The Malaysia Open Science Platform (MOSP) is launched to share research data especially from the public funded research. The platform is connected to institutional repositories and allows access of research data to the public.

2. National Research and Education Network (MYREN)
A network to connect universities in Malaysia.
3. Malaysian Government Central Data Exchange (MyGDX)
A platform to facilitate data sharing between ministries and government agencies.

Institutional level

1. Universiti Malaysia Terengganu (UMT) Open Science Platform
2. OpenData@USM (<https://opendata.usm.my>)
3. Dataverse Repository and Discovery Service
(<https://dataverse.uum.edu.my/>)
4. UM Research Data Repository (<https://researchdata.um.edu.my/>)
5. FRIM Data Center
6. UPM Research Data Repository (<http://putrarepo.upm.edu.my/action.do>)
7. UPSI Digital Repository
8. National Wildlife Forensic Laboratory
9. NIH Data Repository System
10. Malaysian Biodiversity Information System
11. UNIMAS Dataverse (<https://data4research.unimas.my/>)
12. REDCap [Research Electronic Data Capture] (<https://www.project-redcap.org/>)
13. Library open source and cloud computing (<https://library.utar.edu.my/>)

For both levels, most platforms are for exchanges and co-creation of knowledge between scientists and society and/or a federated information technology infrastructure for open science, including high performance computing, cloud computing and data storage. Most of the infrastructure is non-commercial.

14. Does the infrastructure comply with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles

Yes. Many of the infrastructure promotes accessibility to knowledge and the authority of the owner to determine the level of access for each dataset.

15. Does your country host or fund any federated regional or international information technology infrastructure for open science?

- a. Name: Malaysia Open Science Platform (MOSP)
Year: 2023
Language: Malay and English

Target group and beneficiaries: researchers, government, public, industry
Scope: research data sharing from research institutes, higher learning institutes and government agencies categorized by research field.

- b. Name: National Research and Education Network (MYREN)
Year: 2005
Language: Malay and English
Target group and beneficiaries: academia
Scope: MYREN is governed by community agreements under the supervision of the Ministry of Science, Technology, and Innovation (MOSTI) and MYREN's member institutions. These agreements ensure compliance with national research policies and collaboration standards with international NRENs.
- c. Name: Malaysia Open Data Portal (data.gov.my)
Scope: An open data for the government and agencies to share their data and is managed by the National Digital Department under the Ministry of Digital.
Target group and beneficiaries: Public, Researcher, Academia, Government, Industry
- d. Name: Malaysian Biodiversity Information System (MyBIS)
Year: 2016
Target group and beneficiaries: Public, Researcher, Academia, Government, Industry
Scope: one-stop repository database system, which provides and facilitates access to information on biodiversity studies and management in Malaysia

16. Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms?

Yes. This includes:

- Malaysia Open Science Platform
- Open Science Initiatives in Asia Region - Millennium Development Goals (MDG) as well as Sustainable Development Goals (SDG)
- Trans-Eurasia Information Network
- International Science, Technology, and Innovation Centre (ISTIC)

SECTION 4

INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

17. Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your

country. If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

Yes. Capacity building on open science is taking place in Malaysia, at both the national level and institutional level. The capacity building often focuses on current librarians and researchers that are vital stakeholders in the research ecosystem.

Formal Programmes:

Higher education institutions and research bodies conduct structured workshops, seminars, and training programs focusing on open access, FAIR principles, and data management. These programs are often organized in collaboration with MOSTI and the Malaysian Open Science Platform (MOSP) to ensure alignment with national and international Open Science standards. The Data Stewardship on Open Science training is conducted by the Academy of Sciences Malaysia since 2020 and has successfully trained 310 new data stewards of Malaysia. (mosp.gov.my/stewards)

Informal Initiatives:

Informal capacity building includes peer-to-peer learning, knowledge-sharing sessions, and community-driven events within universities and research networks. Collaborations through MYREN and regional networks, such as APAN, facilitate practical experience through hands-on projects and exchange programs.

18. Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country

There is capacity building for policy-makers, however, the exposure is still low. The lack of awareness and underdevelopment of courses is one of the reasons. We acknowledge that there should be more capacity building for policy makers including the legal officers to enhance their awareness on open science and to incorporate open science elements in future policies.

19. Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025?

Yes. Research ethics has been incorporated in the curricular for undergraduate studies for most higher learning institutes in Malaysia. The practice of making research open and accessible is also being taught to the students. However, sharing is still an individual basis.

In addition, open access has been widely accepted in Malaysia where researchers are making efforts to publish their research in an open access journal. Open

access journals are also being published by institutes and organisations in Malaysia.

Through the Malaysia Open Science Platform initiative, more higher learning institutes are developing their own open research repositories and moving towards open science. We will see the rise of data sharing and open science competencies being included in the curricula to adopt the open science values. For example, the integration of Open Science curriculum is expected to be accomplished during the implementation of Open Asia project, participated by Universiti Malaya. <https://openscience.um.edu.my/#>

20. Are open educational resources used as an instrument for open science capacity building in your country in line with the 2019 Recommendation?

Yes. Many institutes in Malaysia are publishing their educational resources on open science in a public platform. This includes institutional platforms, social media, and open mass online open courses (MOOC).

21. Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large?

1. The Academy of Sciences Malaysia is actively promoting open science through social media, roadshows, and engagements with policymakers and research communities. MOSP also encourages experts to join forums, conferences and workshops to promote open science, nationally and internationally.
 - a. Instagram, facebook, X: MOSP_Malaysia
 - b. YouTube: Malaysia Open Science Platform
2. Activities under UMCares (<https://myumcares.um.edu.my/>) and UMSDC (<https://sustainability.um.edu.my/>) are very much aligned with promoting Open Science and Science Communication
3. Science communication by Young Scientist Network, Academy of Science Malaysia.

SECTION 5

FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

22. Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025?

The Malaysian Research Assessment (MyRA) Instrument is a comprehensive system designed to evaluate the research capabilities and performance of all Higher Education Institutions (HEIs) in Malaysia. Introduced in 2006, its primary goal was to support the Malaysian Research University (MRU) initiative by the Ministry of Higher Education (MoHE), which aimed to designate five universities in Malaysia as Malaysian Research Universities (MRUs). Initially, MyRA was used to assess and monitor the research performance of public universities. However, starting in 2014, all HEIs, both public and private, were required to participate in the annual assessment, which coincided with the opening of MoHE research grants to all universities nationwide. MyRA now includes a 6-Star rating system, with all participating HEIs undergoing document and site audits conducted by a team of trained auditors.

There are discussions on implementing open science elements in the research assessment and evaluation process, however, there are still no implemented initiatives relating to the matter.

There are capacity building and awareness activities to shift the mindset to an open assessment and evaluation process. The review of alternative metrics to review the impact of research is being discussed.

23. Have specific clauses that recognize open science practices been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? For example, sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems.

Not currently, but there are some plans towards this initiative is being looked into.

24. Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025?

Yes. There are research awards that include specific categories for research that embraces open science, such as publishing in open access journals and engaging with the community. Institutes also encourage their researchers to publish their findings in open access journals, and those who do often gain recognition in performance evaluation systems.

25. Have there been any unintended negative consequences of open science practices reported in your country? If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

Currently, there are not yet unintended negative consequences of open science reported in Malaysia

SECTION 6

PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 26. Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process?**

Promotion of pre-prints:

Yes.

Exploration of open peer-review practices:

1. <http://www.injet.upm.edu.my/index.php/instruction-to-authors/peer-review>
2. <http://www.pertanika.upm.edu.my/proceeding/peer-review>
3. <https://www.sciencedirect.com/>
4. <https://lib.usm.my/index.php/academicians-researchers/open-science>

Valuing and sharing of negative scientific results and associated data:

Yes, there are awareness activities in promoting the benefits of data sharing

Participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects:

1. Citizen science – UM Water Warriors, engagement with local communities
2. Library with researchers

Development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas:

Stakeholder consultations

Promoting open innovation practices:

1. Open Science, Open Doors: Guide to How Your Research Can Achieve Better Visibility and Impact (20230529161856Current_Evidence_-_Open_Science,_Open_Doors;_A_Guide_To_How_Your_Research_Can_Achieve_Better_Visibility_and_Impact.pdf)
2. Introduction to the idea of a Malaysian Open Innovation Platform that promotes collaboration and openness between the researchers and the industry. The Malaysia Open Science Platform will act as one of the building blocks (data source) for the open innovation platform.

SECTION 7

PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION

IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

27. How is your organisation / institution promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science?

1. Discourse of open science in international platforms such as UNESCO, APEC and Global Open Science Cloud to increase regional and international collaboration.
2. There have been initiatives at other institutions that has collaboration with European universities based on the Erasmus+ project funding to see different practices in Malaysia and other European countries.
3. Open Access Publishing to enhance visibility.
4. Organizes workshops and training on open science.
5. Develop digital platforms for openly sharing data and research outcomes.
6. Encourages researchers to participate in and host international conferences focused on open science, promoting the exchange of ideas and global collaboration.
7. Formation and participation in international networks with other research institutions to facilitate knowledge exchange.
8. Engagement with the focus research groups in the region (Borneo)

28. Does your country or organisation/institution have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science?

1. Malaysia has proposed the formation of the APEC Open Science Alliance (APECOSA) at the APEC Policy Partnership on Science, Technology and Innovation (APEC PPSTI) meeting to promote the sharing of open science best practices between the APEC economies. Following a series of workshops and surveys, the APEC economies agree with the importance of APECOSA to promote scientific advancement towards a sustainable and prosperous region.
2. The Malaysian Academic Library Institutional Repositories - <http://malrep.uum.edu.my/rep/>
3. Universiti Malaya is a member of OPEN ASIA project, which facilitates knowledge exchange of Open Science between EU countries and India. This also includes universities in Malaysia such as Universiti Malaysia Sabah and Universiti Malaysia Sarawak. A forum on open science with all the strategic partners was held at Universiti Malaya on 23 and 24th October 2024. <https://openasia-erasmus.org/>
4. Organizes programs aimed at enhancing the skills of researchers and stakeholders in open science practices.
5. Research Collaboration Projects
6. TEIN (Trans-Eurasia Information Network), SAXS (Southeast Asia Exchange of

Services), EduRoam and GÉANT

7. ASEAN <https://asean.bernama.com/news.php?id=2355101>
8. Participating in networks like APAN (Asia-Pacific Advanced Network), which facilitates collaboration among researchers in the region, and leveraging partnerships with universities and research organizations globally. These efforts focus on sharing best practices in open science methodologies, fostering collaborative research projects, and ensuring that capacity-building initiatives are aligned with international standards.
9. Borneo Satellite Network (BoSaN)
<https://gazette.unimas.my/2024/03/04/zoologists-go-global-by-frst-unimas-successfully-ties-international-collaboration-with-the-faculty-of-forestry-fahutan-universitas-tanjungpura/>
10. Strategic Framework Towards Open Science Adoption in Malaysia
10.1109/TEMSCON-ASPAC52831.2022.9916546

29. Is your country or organisation/institute involved in any discussion on the establishment of regional and international funding mechanisms for open science?

1. UNESCO and Malaysia join forces for Open Science in the Asia-Pacific region <https://mucp-mfit.org/unesco-and-malaysia-join-forces-for-open-science-in-the-asia-pacific-region/>
2. EURAXESS ASEAN speaks on open science at QS Higher Ed Summit: Asia Pacific 2023:
<https://euraxess.ec.europa.eu/worldwide/asean/news/euraxess-asean-speaks-open-science-qs-higher-ed-summit-asia-pacific-2023>
3. ASEAN Framework on Science, Technology, and Innovation
4. <https://cdo.unimas.my/open-asia/about-project>
5. Dissemination of information on international funders are circulated at national level

SECTION 8

GENERAL CONSIDERATIONS

30. Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country or organisation/institution

There are several external factors that could impede the implementation of the 2021 UNESCO Recommendation on Open Science in Malaysia. Key challenges include limited funding for open science initiatives, which affects the ability of institutions to adopt necessary practices and technologies. Additionally, infrastructural issues, such as insufficient access to digital platforms for data sharing and collaboration, hinder progress. Cultural resistance within traditional

research settings can also pose challenges, as researchers may be hesitant to adopt open science practices due to concerns over intellectual property and the perceived risks associated with public data sharing. Addressing these challenges requires coordinated efforts from government bodies, academic institutions, and international partners to promote the benefits of open science and develop supportive policies and infrastructure.

Prepared by:

Knowledge Hub, Data Analytics and Strategic Inputs Unit,
Academy of Sciences Malaysia
2 December 2024

Endorsed by:



Name: Professor Ir Dr Shaliza Ibrahim
FASc
Position: Co-Chair of Malaysia Open
Science Alliance
Date: 10/12/2024



Name: Professor Dr ChM Noorsaadah
Abd Rahman FASc
Position: Co-Chair of Malaysia Open
Science Alliance
Date: 10/12/2024